Chen et al. 2020 — Nasopharyngeal Carcinoma · 10x · 3,236 malignant cells (8 samples)

Do malignant cells recapitulate embryonic-development programs? (Gast / pSom / Crest on a cell-state manifold)

GATE 1 (mixing + continuity) GO

8/14 samples pass min-cell threshold 107; Harmony(sample,theta=2) fully intersperses all 8 patients; 1 connected component after island removal (438 detached cells dropped). Structure is mixed and continuous.

GATE 2 (embryonic module geography) NO-GO

A robust Crest-terminal axis exists (Crest opposes Gast -0.64 / pSom -0.44, SURVIVES cell-cycle correction -> not a proliferation artifact). But the defining embryonic BODY is absent: Gast/pSom are decoupled (r=+0.01 vs Kurten GO +0.33 / Puram GO -0.49) and pSom does not participate (r(t,pSom)=-0.17, flat). A two-module Gast/Crest gradient without a coupled Gast/pSom body is not the embryonic pattern.

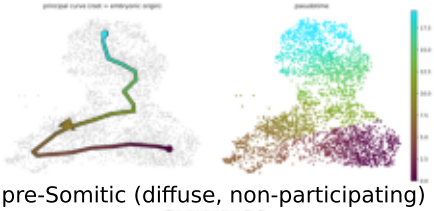
GATE 3 (cross-repression) NA

No bifurcation on a linear manifold -> no Gast<->pSom arms to test. Correlation + modules notebooks not produced.

Preprocessing: Harmony(sample)-integrated UMAP



Manifold: non-branching curve + pseudotime (Gate 1 GO)



Gastrulation (early end of axis)



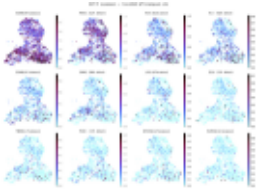
pre-Somitic (diffuse, non-participating)



Neural Crest (terminal inverse gradient)



EMT-TFs (detection-screened)



Key parameters (all data-derived, shown in-notebook)

Expression	UMI counts -> normalize_total, log1p (no scale)
Threshold	min 107 malignant cells/sample (median-restricted gap 36->107, x2.97) -> 8 samples, 3,674 cells
Embedding	overdispersed genes -> PCA(19 PCs, each >0.5% var) -> Harmony(sample,theta=2) -> knn15 -> UMAP -> Leiden res=0.3
Island removal	438 cells (clusters 2,4,6,7,8) dropped -> 3,236 cells, 1 connected component
Subtypes	Cycling (c0/c1), Protein-maturation (c3), Hypoxia (c5); markers + 3CA meta-programs (mp_top) + Chen malignant calls
Manifold	scFates curve on X_umap (non-branching; 0 branch nodes across 12-combo PPT sweep); rooted Gast/pSom-high tip
EMT-TFs shown	SOX4, PRRX1, TCF4, TCF3, SOX9, SNAI2, LEF1, ZEB2, TBX3, FOXC1, NFATC2, RUNX2 (all 12 with detect>=10%); canonical SNAI1/ZEB1/TWIST1 near-absent, dropped

VERDICT — NOT REPRODUCED (embryonic geography absent; Gate 2 NO-GO). Malignant NPC cells form a mixed, continuous, non-branching manifold with a real, cell-cycle-independent Crest-terminal gradient (Crest opposes Gast -0.64 / pSom -0.44 after cc-correction). But the three-module embryonic pattern is not present: Gast/pSom are decoupled (r=+0.01, vs Kurten GO +0.33) and pSom does not participate in the axis (flat, r(t,pSom)=-0.17). NOTE: this NO-GO corrects, rather than echoes, the prior attempt's 'generic-carcinoma' rationale — the axis is embryonic-flavoured and NOT mere proliferation; what fails is the coupled Gast/pSom body the gate requires.